Definition 20.1

A *homomorphism* from a ring R to a ring S is a function $f\colon R \to S$ such that for any $a, b \in R$ we have

- $f(a+b) = f(a) + f(b)$
- $f(ab) = f(a) \cdot f(b)$

Definition 20.2

A *isomorphism* of rings is a homomorphism $f\colon R \to S$ which is a bijection.

If there exists an isomorphism between rings R and S then we say that these rings are *isomorphic* and we write $R \cong S$.

Theorem 20.3

If $f\colon R \to S$ is an isomorphism of rings then the inverse function $f^{-1}\colon S \to R$ is also an isomorphism of rings.

Definition 20.4

Let $f\colon R \to S$ be a homomorphism of rings. The *image* of f is the set $\mathrm{Im}(f) \subseteq S$ defined by

$$\mathrm{Im}(f) = \{f(r) \mid r \in R\}$$

The *kernel* of f is the set $\mathrm{Ker}(f) \subseteq R$ given by

$$\mathrm{Ker}(f) = \{r \in R \mid f(r) = 0\}$$

Theorem 20.5

Let $f\colon R \to S$ be homomorphism of rings. Then

1) $\mathrm{Im}(f)$ is a subring of S

2) $\mathrm{Ker}(f)$ is an ideal of R.

Theorem 20.6 (First Isomorphism Theorem for Rings)

Let $f\colon R \to S$ be a ring which is onto. Then $S \cong R/\mathrm{Ker}(f)$.

Theorem 20.7

If R is a ring and $I \triangleleft R$ then there exists a ring homomorphism $f\colon R \to S$ such that $\mathrm{Ker}(f) = I$.